

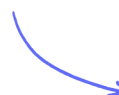
## Multiple Choice Questions

# Intermolecular Forces

Intermolecular Forces - Introduction / Hydrogen Bonding / Physical Properties / Solvent Choice

|                      |            |
|----------------------|------------|
| Easy (3 questions)   | /4         |
| Medium (7 questions) | /7         |
| <b>Total Marks</b>   | <b>/11</b> |

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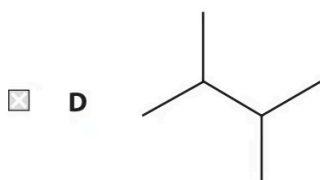
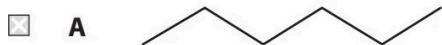
# Easy Questions

1 Which of these pure compounds has hydrogen bonding in the liquid state?

- A. 1,1,1-trichloroethane,  $\text{CH}_3\text{CCl}_3$
- B. trimethylamine,  $(\text{CH}_3)_3\text{N}$
- C. hydrogen fluoride,  $\text{HF}$
- D. hydrogen sulfide,  $\text{H}_2\text{S}$

(1 mark)

2 Which of these isomers has the **highest** boiling temperature?



(1 mark)

3 (a) This question is about alkanes.

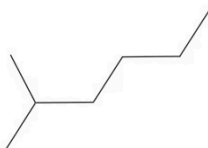
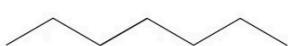
Which of these alkanes has the **highest** boiling temperature?

- A. butane
- B. hexane
- C. pentane
- D. propane

(1 mark)

(b) Which of these alkanes has the **lowest** boiling temperature?

(1)

- ☐ A 
- ☐ B 
- ☐ C 
- ☐ D 

(1 mark)

# Medium Questions

1 Which of these isomers with the formula  $C_8H_{18}$  has the lowest boiling temperature?

- A.  $CH_3CH(CH_3)CH_2CH_2CH(CH_3)CH_3$
- B.  $CH_3CH(CH_3)CH_2CH_2CH_2CH_2CH_3$
- C.  $CH_3C(CH_3)_2CH_2CH(CH_3)CH_3$
- D.  $CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

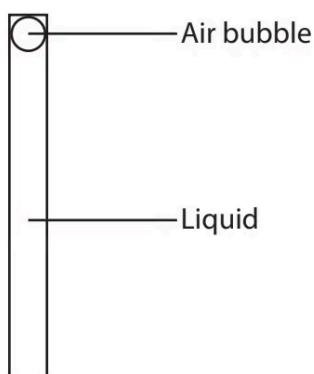
(1 mark)

2 Which sequence shows the molecules in order of **increasing** boiling temperature?

- A.  $H_2O < Br_2 < Cl_2 < CH_4$
- B.  $Br_2 < CH_4 < Cl_2 < H_2O$
- C.  $Cl_2 < CH_4 < H_2O < Br_2$
- D.  $CH_4 < Cl_2 < Br_2 < H_2O$

(1 mark)

3 The viscosities of four liquid organic compounds are compared by placing the liquids in separate identical tubes, each with an air bubble at the top.



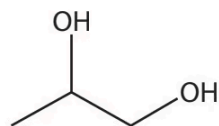
The tubes are inverted, and the times measured for the air bubble to travel the length of

the tube. For which compound does the air bubble take the longest time?

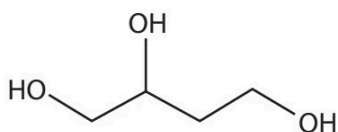
☐ **A**



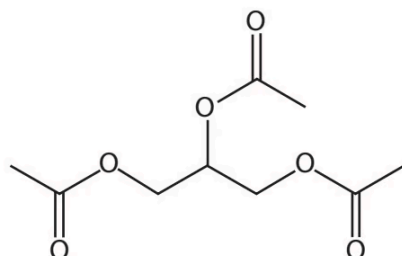
☐ **B**



☐ **C**



☐ **D**



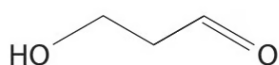
(1 mark)

4 Which is **not** correct about ice?

- A.** ice has a lower density than water
- B.**  $\text{H}_2\text{O}$  molecules are further apart in ice than in water
- C.** the  $\text{H}-\text{O}-\text{H}$  bond angle is the same in ice and in water
- D.**  $\text{H}_2\text{O}$  molecules in ice are held together by hydrogen bonds

(1 mark)

5 Which intermolecular forces exist **between** the molecules of the compound shown?



- A.** hydrogen bonding and London forces only
  - B.** hydrogen bonding and permanent dipole-permanent dipole forces only
  - C.** London forces and permanent dipole-permanent dipole forces only
  - D.** hydrogen bonding, permanent dipole-permanent dipole forces and London forces
- (1 mark)**

**6** Which solvent dissolves the greatest amount of hydrocarbon  $C_{35}H_{72}$ ?

- A.** butan-1-ol
- B.** ethanoic acid
- C.** hexane
- D.** water

**(1 mark)**

**7** Which row shows the hydrogen halides in order of **increasing** boiling temperature?

|                          |          | <div style="display: flex; align-items: center; justify-content: space-between;"> <div>Lowest</div> <div>→</div> <div>Highest</div> </div> |     |     |     |
|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|
| <input type="checkbox"/> | <b>A</b> | HF                                                                                                                                         | HCl | HBr | HI  |
| <input type="checkbox"/> | <b>B</b> | HI                                                                                                                                         | HBr | HCl | HF  |
| <input type="checkbox"/> | <b>C</b> | HCl                                                                                                                                        | HBr | HI  | HF  |
| <input type="checkbox"/> | <b>D</b> | HF                                                                                                                                         | HI  | HBr | HCl |

**(1 mark)**